RECOMMENDED EXPERIENCE TYPE

AI Lab

Check It: Cronbach's Alpha Analysis is a reliability test that checks whether multiple survey questions consistently measure the same underlying concept

Why this type: You want students leaving able to tell strong evidence from weak evidence; competence in this field looks like investigating; you want them holding a conclusion with its evidence.

AI Lab means participants use AI to research, verify, and build an evidence-based conclusion.

LENGTH
50 minutes · 5 phases

TEAMS
4 students, 4 roles

AI VISIBILITY
Embedded (Lane 2 — in class)

DOMINANT LITERACY
Critiquing AI and Recognizing AI

DOMINANT PILLAR
Pillar 3 — Critical Evaluation

READ THESE FIRST

SCOPE

The topic reads like more than one unit. Cut it to the single thing students most often get wrong. The eighty-percent test is the bar: can eight in ten teams finish an artifact in 50 minutes even if something goes sideways?

PROTECT THE CLEANING STEP

Cleaning is where the judgment lives. Give it five explicit minutes and do not hand teams a pre-cleaned file — the session gets thinner the tidier the data is.

ADULT LEARNER TEST

This is new territory. Substitute close-to-home experience: ask what they have seen as a patient, customer, parent, or employee rather than as a practitioner.

How the four types scored

Your answers point most strongly at one type, and the runner-up is a real option rather than a leftover. Both framings teach a transferable AI skill.

AI Lab	9.0
AI Quests	3.5
AI Studio	0.0
AI Arena	-1.5

THE RUNNER-UP — AI QUESTS

Run it as Quests if you want students building something a colleague could walk through — depth of understanding becomes visible in the build.

TWO RULES THAT OVERRIDE ANY SCORE

No real clock, no Arena. A countdown on a critique activity is decoration, not a mechanic — that session is really a Lab. **And if the artifact is for someone else to move through, it is Quests, not Studio**: Studio's artifact is for the maker, Quests' artifact is for a learner.

THE ACTIVITY

What students actually do

Teams take five confident-sounding claims about Cronbach's Alpha Analysis is a reliability test that checks whether multiple survey questions consistently measure the same underlying concept and test each one against the dataset.

How it runs

1. Teams start with the raw data on Cronbach's Alpha Analysis is a reliability test that checks whether multiple survey questions consistently measure the same underlying concept — working out what it actually measures and what has to be cleaned before it can be used at all.
2. AI supplies support in seconds — and at least one piece of it will not say what it was cited for. Teams find that themselves; nobody points at it.
3. Every claim ends up marked holds, fails, or cannot tell — and the third category turns out to be the interesting one.

They leave with

A marked claim set on Cronbach's Alpha Analysis is a reliability test that checks whether multiple survey questions consistently measure the same underlying concept: five claims, each with a verdict and what in the dataset supports or breaks it.

NOT THE ONE YOU PICTURED?

Three activities fit AI Lab. They teach the same skill, run on the same phases and timings, and hit the same wow and failure moments — only what students make changes. All three work with the dataset.

The claim audit ✓ CHOSEN

Teams take five confident-sounding claims about Cronbach's Alpha Analysis is a reliability test that checks whether multiple survey questions consistently measure the same underlying concept and test each one against the dataset.

The sourced brief FITS THE DATASET

One genuinely contested question about Cronbach's Alpha Analysis is a reliability test that checks whether multiple survey questions consistently measure the same underlying concept, answered in a page from the dataset, with the evidence trail attached and the gaps admitted.

Trace it back FITS THE DATASET

Teams take a statement about Cronbach's Alpha Analysis is a reliability test that checks whether multiple survey questions consistently measure the same underlying concept that everybody repeats, and work back through the dataset to find where it actually came from.

CONCEPT CARD

The session, in one page

COURSE

Marketing Research

TOPIC

Cronbach's Alpha Analysis is a reliability test that checks whether multiple survey questions consistently measure the same underlying concept

WHAT HAPPENS NOW

I would show students a concrete example and let students discover the need for the statistic.

RAW MATERIAL

A dataset, results, or records. Phase 1 opens on it: Open the dataset. Before anything else, name what each field actually measures, what is missing or miscoded, and what you would have to clean before trusting one conclusion from it.

TRANSFERABLE SKILL

Making an AI show its evidence, then checking whether the evidence actually supports the claim.

ARTIFACT

A marked claim set on Cronbach's Alpha Analysis is a reliability test that checks whether multiple survey questions consistently measure the same underlying concept: five claims, each with a verdict and what in the dataset supports or breaks it.

WOW MOMENT

In under two minutes the AI returns an organised sweep of positions and sources on the topic — an evening of reading, compressed. The wow is the speed; the lesson lands when the checking starts.

FAILURE BY DESIGN

The AI will state something confidently that is wrong, unsourced, or flattened past the point of truth. Students must find it by verifying, not be told. If nobody catches it by the halfway mark, ask the room one question: which of these claims would you sign your name to?

PLANT THIS SPECIFIC ERROR

Using Cronbach's Alpha on a scale that measures multiple things and reporting a single alpha for the entire instrument.

HUMAN CONTRIBUTION

The judgment call on evidence: deciding whether a source is credible and whether it actually supports the claim it is attached to. Only the student can make that call, and the brief cannot be finished without it.

ENTRY POINT

Required: a laptop and the topic you already teach. Not required: any coding, any prior AI experience, any paid tool or account. Students use an institutionally approved AI tool of their choice.

SHARE FORMAT

State the conclusion and its evidence, then the one claim AI got confidently wrong and how it was caught. Works in a virtual classroom: screen-share plus one line in chat.

PILLAR 2 — SAFE USE

Approved tools only. No real patient, client, student or employee data goes into any AI tool. If a student wants to use a non-approved tool, they pair with someone who is on an approved one.

PILLAR 4 — ATTRIBUTION

The artifact carries one line: what AI produced, and what the team decided. Reflection question 3 asks for it out loud, in every version of this session.

Phases

#	PHASE	MIN	WHAT STUDENTS DO	UNLOCK WORD
1	**Set the question**	8	Open the dataset. Before anything else, name what each field actually measures, what is missing or miscoded, and what you would have to clean before trusting one conclusion from it.	—
2	**Fast sweep**	10	Write five flat, confident claims about Cronbach's Alpha Analysis is a reliability test that checks whether multiple survey questions consistently measure the same underlying concept, then use AI to find support for each. Speed is the point here.	**sweep**
3	**Audit the claims**	14	Check every piece of support against the dataset. Does it say what it was cited for?	**audit**
4	**Build the evidence trail**	10	Mark each claim holds / fails / cannot tell, and write one line on why.	**trail**
5	**Present and reflect**	8	Present one claim that failed and how you caught it.	**present**

Roles — teams of 4

1. **Question keeper** — owns the written claim under test, in one sentence
2. **Prompter** — owns the prompt log: every prompt sent, in order
3. **Verifier** — owns the source list and the verdict on each claim
4. **Reporter** — owns the finished brief and its evidence trail

AI skill moments — what to watch for

WHEN IT HAPPENS	THE SKILL	WHAT YOU DO
Phase 2, when a team accepts a claim with no source attached	**Prompting for evidence rather than assertion**	Ask them to re-prompt for the source of that specific sentence. Let the room hear what comes back.
Phase 3, the first time a source turns out not to say what was claimed	**Checking the citation, not just its existence**	Put it on the projector. This is the highest-value sixty seconds in the session.
Phase 3, when a team treats a confident tone as a credential	**Separating confidence from accuracy**	Ask: how sure does it sound, and how sure should you be? Name the gap out loud.

Reflection

1. Where was the AI confident and wrong, and what made you check that one?
2. What is the difference between how fast it answered and how reliable it was?
3. Name what the AI produced and what you decided. Which of the two mattered more to how this turned out?
4. Where in your own work or study, in the next two weeks, could you use this exact move?

> *Closing stem, everyone completes it: "The most useful thing I learned about working with AI today was ___, and I could use this when ___."*

SAY THIS AT THE CLOSE

A high Cronbach's Alpha doesn't prove you're measuring the right thing. It only suggests your questions are measuring the same thing.

Pacing

If time runs short: halve the "Build the evidence trail" phase and cut sharing to three volunteers, everyone else submits one line. Never cut phase 2 ("Fast sweep") or the reflection — the wow moment and the skill live there.

If energy is high: teams trade briefs and try to break each other's conclusion — a five-minute adversarial pass.